

How Autocracy Shapes the Social Sciences

Synthesized Evidence from an Agent Orchestra

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Does autocracy bend the *content* of social science?

- Autocracies are now major science producers (China > US).
- Old claim: **open societies** breed good science (Popper, 1945; Kitcher, 2001).
- Bibliometrics shows regime effects on **volume** and **citations** (Tayebi and Teimouri, 2025; Yang and Liu, 2021).

The gap

If researchers self-censor, autocracy should also shape the **contents and framing** of research, not just its quantity. We look at this in Social Sciences and Humanities (SSH) field.

The project

Research questions

1. Do researchers in autocracies **self-censor** and respond strategically to regime context?
2. What are the implications for **social-science contents**?

Design

- An orchestra of AI-agents, to test several implications
- 25 **pre-registered** AI-agent teams
- One corpus: **2.7M** WoS SSH articles (1970–2023)
- Autocracy-democracy = V-Dem liberal democracy
- 5 sub-theory families, family-wise Bonferroni corrections

So far (21/25 teams)

- **Essentially no signal of self-censorship theory**
- 0 of 21 significant
- Method: an **agent-orchestra** many-analysts design

Theory: strategic self-censorship

Researchers and their options.

- Driven by **career** (prominence, tenure, funding) and **scientific** goals (move the field, solve puzzles).
- But also value physical, economic, and social **security**, and act strategically within regime-constraints
- Rationally to regime **sticks** (sanction, lost funding) and **carrots** (state-controlled research money).
- Strategic margins: *what* to study, *how* to frame it, *who* to collaborate with, how *visible* to be.

No single decisive test.

- Regimes' "red lines" are uncertain, to rulers and researchers alike.
- Several substitutable levers to pull (framing, contents, collaboration, language, etc.)
- Self-censorship therefore surfaces as **many weak signals**, spread across domains and dependent on context.

Test many implications on many outcomes, not one.

Five families of observable implications

Topic avoidance	avoid sensitive subjects, keywords, fields.	7
Framing neutrality	hedge, depoliticize, go technocratic.	6
Collaboration constraint	fewer international ties with democracies.	5
Visibility suppression	censored work gets cited less.	4
Ideological alignment	regime-consonant content.	3

Each → a prediction on Liberal democracy.

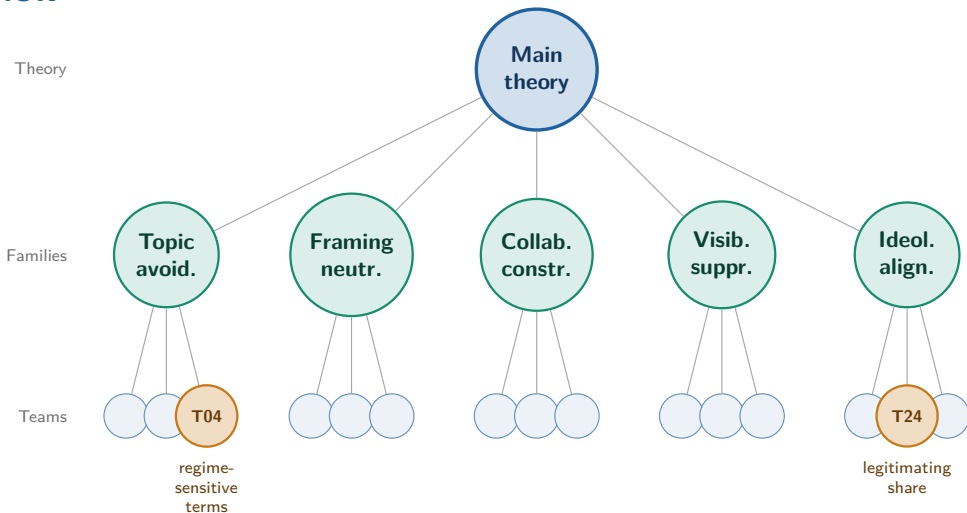
Pre-registration

- One theory, ~ 25 operationalizations, one corpus.
- Avoid **researcher degrees of freedom** and cherry-picking

Response

Pre-register **many** designs, hypotheses, lock them before results, and read the whole **distribution** of estimates.

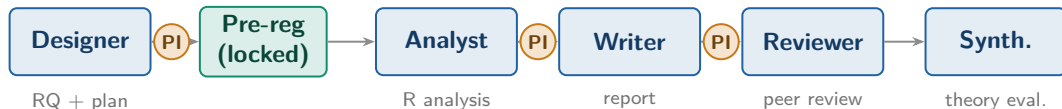
Overview



teams tests offspring implication; the theory is judged on the *distribution* of results.

The Agent Orchestra

25 teams, each a pipeline of AI agents, with a PI gate at every stage (human in the loop):



Amber: PI review gate (redesign / revise loop). Hypotheses locked before analysis at Git 960525f; an emerging method (Zhu and Wang, 2026; Gupta et al., 2026).

All 25 teams: outcomes and status

Topic avoidance

- 01 PCI (political-content index)
- 02 Keyword entropy
- 03 Share in sensitive fields
- 04 Share w/ regime-sensitive terms
- 05 Share critical of own governance[†]
- 06 Semantic diversity (embeddings)[†]
- 07 Share internationally funded

Framing neutrality

- 08 Epistemic hedging rate
- 09 Normative language rate
- 10 Share purely technocratic[†]
- 11 Argumentative stance rate
- 12 Share w/ normative conclusion[†]
- 13 Share in domestic journals

Collaboration constraint

- 14 Mean co-author countries
- 15 Share democratic co-authors
- 16 Share domestic-only authorship
- 17 Mean author count
- 18 Mean distinct institutions

Visibility suppression

- 19 Log citation ratio
- 20 Citation gap by sensitivity
- 21 Share ever cited
- 22 Citation Gini

Ideological alignment

- 23 Share liberal/neutral framing
- 24 Share legitimating regime
- 25 Share anti-liberal framing

[†] pending (LLM / embeddings); the other 21 are complete.

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Data: one locked corpus

Corpus (Web of Science, SSH)

- 2.7M articles, 167 countries
- 1970–2023
- 99.98% matched to V-Dem

Politics (V-Dem)

- IV: v2x_libdem
- Controls: log GDPpc, log pop

Unit: **country–year**.

One shared benchmark specification

Two-way fixed effects, every team:

$$Y_{it} = \beta \text{v2x_libdem}_{it} + \gamma_1 \log \text{gdppc}_{it} + \gamma_2 \log \text{pop}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

- Country + year FE; SE clustered by country.
- Secondary: pooled OLS. Robustness: alt. measure + binary regime.
- **Family-wise Bonferroni**, $\alpha = 0.05/k$.

Example 1, Team 04: topic avoidance

Rationale. Regime-sensitive concepts such as democracy, human rights, corruption, and protest are the highest-cost topics under autocracy. A paper's most visible labels (title, keywords) are the likeliest site of **anticipatory avoidance**.

- **Outcome** `share_regime_sensitive`: share of a country-year's articles with ≥ 1 hit from a regime-sensitive dictionary (democracy, rights, corruption, protest, repression, ...) in title, keywords, or abstract.
- **Test:** TWFE on `v2x_libdem`; **predicted** $\beta > 0$ (more democracy $\rightarrow$ more sensitive terms).

Result

$\hat{\beta} = +0.001$ (SE 0.018), $p = 0.94$, $N = 2,803$ country-years.

Indistinguishable from zero.

Example 2, Team 24: ideological alignment

Rationale. Autocracies do not only suppress dissent; **reward** scholarship that endorses the regime (state funding, party-controlled outlets, promotion criteria).

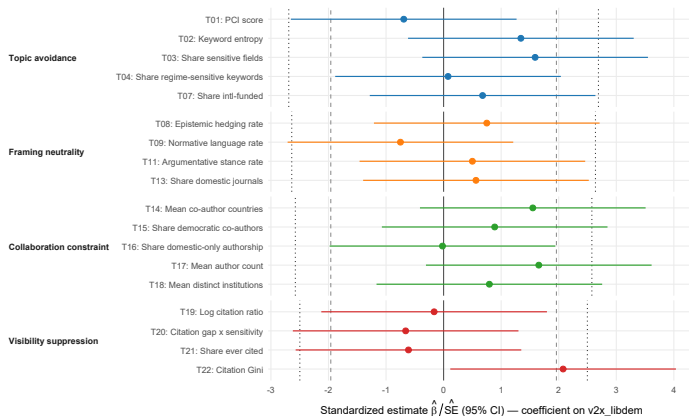
- **Outcome** share_legitimizing: share of abstracts an **LLM** (Claude Haiku) labels as actively endorsing the political system; stratified sample, ~141k abstracts classified.
- **Test:** TWFE on v2x_libdem; **predicted** $\beta < 0$ (more autocracy $\rightarrow$ more legitimation).

Result

$\hat{\beta} = +0.003$ (SE 0.002), $p = 0.23$, $N = 3,207$ country-years. **Wrong sign, not significant.**

Only 582 of 140,879 classified abstracts (0.4%) coded as legitimating: a rare-event outcome. Completed after the May-27 forest plot was frozen, not shown there.

Results: coefficients cluster at zero



Each row = one pre-registered team test (TWFE: country and year FE, GDP/pop controls, clustered by country).
 Solid line: zero. Dashed: nominal ± 1.96 . Dotted: family-specific Bonferroni 5% threshold.
 Sign convention: positive z = outcome rises with liberal democracy (typically the theory-predicted direction).
 16 of 25 pre-registered teams reported here; 7 LLM/embedding-based teams (05, 06, 10, 12, 23–25) pending.

Preliminary: 21 of 25 teams done; 4 framing/embedding teams pending.

Nothing survives correction

Sub-family	Registered (k)	Completed tests	Sign of $\hat{\beta}$		Nominal $p < .05$	Bonferroni $p < .05$
			> 0	< 0		
Topic avoidance	7	5	4	1	0	0
Framing neutrality	6	4	3	1	0	0
Collaboration constraint	5	5	4	1	0	0
Visibility suppression	4	4	1	3	1	0
Total	22	18	12	6	1	0

Four corpus-based families (18 teams) shown.

- **0 of 18** survive Bonferroni; 1 nominal hit (T22) runs **opposite** to prediction.
- Ideological-alignment teams **23–25** (now complete) are **also null**: $p = 0.13, 0.23, 0.16$.
- Our two examples land here too: **T04** $p = 0.94$, **T24** $p = 0.23$.

Results

Robustness

- Within-family directions suggestive at best; none significant.
- TWFE and pooled OLS **agree**: not a fixed-effects artifact.

Power, the leading alternative explanation

- Large samples (2,500–3,500 country-years, 150+ countries): well-powered for **between-country** contrasts.
- But TWFE leans on **within-country** change in democracy, which is slow and rare, so the within estimand is less precisely identified.
- Reassurance: confidence intervals sit **tight around zero**, ruling out large effects, and the cross-sectional (pooled) estimates agree.
- **Still open**: small effects, or effects concentrated in specific regimes, fields, or periods, cannot be excluded.

Results

- Across 21 dimensions, including LLM-coded **ideological alignment**, **little sign** that autocracy bends published content.
- **Caveats:**
 - Observational; content censorship vs. **venue selection**.
 - WoS skews English / high-impact; misses domestic outlets.
 - **4 framing teams** (LLM, embeddings) still pending.

Preliminary: 21 of 25 teams done; 4 framing/embedding teams pending.

Summary and discussion

Findings. Across 21 pre-registered tests, including LLM-coded ideological alignment, **little evidence** that autocracy bends the content of published SSH; no coefficient survives correction.

Contribution

- *Theoretical:* a first broad test of content self-censorship; an informative null that shifts the burden of proof.
- *Methodological:* an **agent-orchestra** many-analysts design that scales a multi-implication theory, with pre-registration and human review built in.

Limitations. Observational (content censorship vs. venue selection); WoS skews English and high-impact, missing domestic outlets; effective power for the within estimand is limited; 4 framing teams pending; AI-generated design choices raise provenance and refereeing questions.

► [Appendix / backup slides](#)

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Appendix: backup slides

- ▶ All 25 teams: outcomes and status
- ▶ Five families and Bonferroni thresholds
- ▶ Estimation: specification and robustness
- ▶ Data and corpus construction
- ▶ LLM classification (ideological alignment)
- ▶ Pre-registration, gates, pending teams

Five families and Bonferroni thresholds

Sub-family	Teams	k	Adjusted α ($0.05/k$)
Topic avoidance	01–07	7	$p < 0.0071$
Framing neutrality	08–13	6	$p < 0.0083$
Collaboration constraint	14–18	5	$p < 0.0100$
Visibility suppression	19–22	4	$p < 0.0125$
Ideological alignment	23–25	3	$p < 0.0167$
Total	01–25	25	

Bonferroni applied within each family; predicted signs are in each team's preregistration.

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Estimation: specification and robustness

Primary (TWFE), every team:

$$Y_{it} = \beta \text{v2x_libdem}_{it} + \gamma_1 \log \text{gdppc}_{it} + \gamma_2 \log \text{pop}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

- α_i country FE, δ_t year FE; SE clustered by country.
- **Secondary:** pooled OLS, year FE only (cross-sectional pattern).
- **RC1:** alternative operationalization or sample restriction.
- **RC2:** replace `v2x_libdem` with `lied_binary`.
- Unit: country-year; minimum 10 articles per country-year (most teams).

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Data and corpus construction

- Source: Web of Science, all SSH subject categories.
- 3,189,557 article-country rows; 2,709,224 unique articles.
- Years 1970–2023; 167 countries; 99.98% matched to V-Dem by ISO3.
- De-duplication and V-Dem merge: `scripts/00_prepare_data.R`.
- IV `v2x_libdem` (0–1); controls log GDP per capita, log population.
- Robustness IV `lied_binary`; regime type `v2x_regime` (0–3).
- Corpus locked before any analysis.

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LLM classification (ideological alignment)

Teams 23–25 code abstracts with an LLM (Claude Haiku). Example, Team 24:

- **Binary label.** LEGITIMATING: the abstract explicitly endorses or validates the political system, state authority, ruling party, or leadership as effective, beneficial, or necessary. NON-LEGITIMATING: everything else (default when in doubt).
- **Sample.** Stratified by country $\times$ regime $\times$ decade; up to 10 abstracts per stratum; abstracts of ≥ 50 characters; fixed seed.
- **Aggregation.** Share LEGITIMATING per country-year, then TWFE as usual.
- **Scale.** About 141k abstracts classified for Team 24.

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Pre-registration, gates, pending teams

Pre-registration. All 25 hypotheses, predicted signs, primary specifications, robustness checks, and Bonferroni families were locked at Git commit 960525f (6 May 2026), before any analyst session ran.

- **Gates.** PI review at design (B), results (D), and report (G); revision loops allowed for implementation only, not for outcomes.
- **Complete (21):** all corpus-based teams plus ideological alignment (23–25).
- **Pending (4):** 05 (critical governance, LLM), 06 (semantic diversity, embeddings), 10 (technocratic share, LLM), 12 (normative conclusion, LLM).

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